Alexandra Heneghan

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Portfolio

EDUCATION

University of Colorado Boulder | B.S. Aerospace Engineering, Minor in Creative Technology & Design | **May 2028**

Technical Skills :

Programming & data analysis: MATLAB, C++, R, Microsoft Excel, Arduino IDE, ESP32

Design & Prototyping: TouchDesigner, Blender, DMX Lighting, UI Integration

AI & Development: Claude, Codex, Prompt Engineering, AI Integration, MCP

PROJECTS

Safety Bracelet | Chief Technology Officer (CTO) **Boulder, Colorado** *2026 - present*

- Initiated development of a wearable alcohol safety device designed to estimate BAC through sweat analysis, researching sensor integration and technical feasibility.
- Developed digital prototypes and designed the initial user interface, collaborating with a six-member team to refine product functionality and design.
- Conducted market research, customer profiling, competitive analysis, and financial forecasting to guide product strategy.

Light Tower Project | Interactive Technology & Design Team Member **Boulder, Colorado** *2025*

- Collaborated with a nine-member team and building architects to design an interactive lighting installation for CU Boulder's ATLAS Light Tower.
- Programmed DMX-controlled lighting sequences and developed an audiovisual show synchronized to the *Interstellar* soundtrack for a final show.

Glider Design Project | Engineering Team Member **Boulder, Colorado** *2025*

- Designed, built, and flight-tested a hand-launched glider using CAD and MATLAB to analyze aerodynamics, center of gravity, and flight stability.

Whiffle Tree Experiment | Engineering Team Member **Boulder, Colorado** *2024*

- Conducted wind tunnel testing of a Clark Y-14 airfoil to analyze lift and drag, and used MATLAB to evaluate structural responses under varying load distributions.

Personal Projects : ESP32 Animated Display, TouchDesigner Animation, App Development

LEADERSHIP EXPERIENCE & ACTIVITIES

Engineering Peer Leader - Mentor

August 2025 - 2026

- Supported first-year engineering students in transitioning to college through academic and personal mentorship.
- Co-lead weekly 50-minute seminar sessions, leading discussions on engineering ethics, philosophy (Kant and other foundational thinkers), and how these ideas connect to students' future roles as engineers.

Microsoft Copilot Ambassador *2026 - present*

- Promoted Microsoft Copilot through student outreach and campus engagement, demonstrating AI-powered tools and their applications in academic and professional workflows.